Exp. No.: 4	**Determining the Tool Center Point**
Date:	

Aim:

To determine the Tool Center Point (TCP) using automatic calibration in RT Toolbox3.

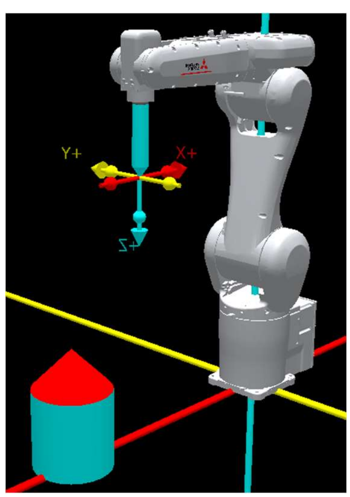
Procedure:

- Power On and Initialize the Robot
- Command the robot to move to its designated home position.
- Attach a specific tool for to the robot's end-effector for determining the TCP.
- Orient the tool to ensure correct positioning with a workpiece.
- Select Tool Automatic Calculation in Robot name>Tool>Tool Automatic Calculation> Auxiliary point> 1 Point and record the position using Teach Selection Line.
- Repeat the two above steps in two additional orientations for the tool.
- Note the calculated tool coordinate and Presumed Error.
- Rotate the tool about the three axes at the calibrated point to visually verify the calibration.
- Increase the number of points for Teach Selection Line for decreasing the error and write into the controller.
- Send the robot back to its home position once the calibration is finished.

Teach Pendant Mode:

Enable TB enable switch
Pull the lever on the backside of the pendant
Change Override to <20
Turn on Servos
Orient tool > 1 Point; Orient tool > 2 Point; Orient tool > 3 Point
Perform Tool Automatic Calculation and record error

Simulation:



Hand details

Base = 20,20,150 position =0,0,0

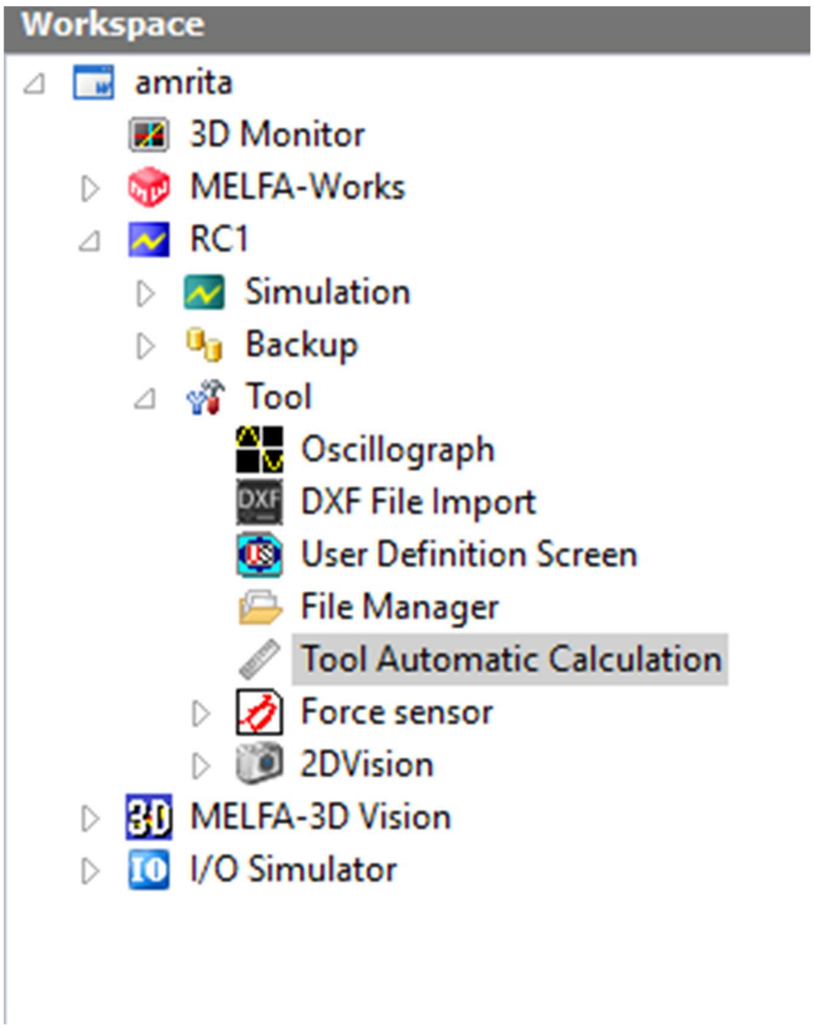
Cone = 3,20,25 position = 0,0,150

Tip =0,0,175 position=0,0,175

Base cylinder and cone details

Cylinder=100,100,200 position =700,0,0

Cone=1,100,100 position =700,0,200



Tool automatic calculation

Capture multiple points in tool automatic calculation to get tool center point for the pen hand and object kept in front

Observation:

- Correlate the error with the deviation in the Tool Center Point and verify that the error decreases by using more points for calibration.

Results & Inference: